

Essentialism and the Essence of *to be Necessary*

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Fine (1994) argued against the view that essence should be understood in terms of *de re* modality. He suggested instead to understand metaphysical necessity by essence of things. The idea to explain, especially to ground all metaphysical necessity in essence was followed by various philosophers albeit with some differences in details. I term all such accounts *essentialism* on necessity.

But if essence is not to be understood by modal notions, one might ask what makes essence so intimately related to necessity. Especially, some philosopher argued that essentialists cannot account for the reason why *to be essential* is such that all facts obtaining essentially are necessary. If essence is partially grounded on necessity, essentialism infringes the asymmetry of ground; but if essence is non-modal, then there is a conceptual gap between what is it to obtain essentially, and what is it to be necessary.

In this paper I argue that there is no conceptual gap between non-modal conception of essence and necessity, if one takes metaphysical necessity is the necessity which is essentially essentialist. In particular, the extensional equivalence between facts which obtain essentially and are metaphysically necessary can be taken as obtaining essentially of *to be necessary*. I also argue that this move is forced upon essentialists.

After some clarification on terminology, §2 will be devoted to elaborate the challenge of gap, particularly raised against essentialists. In §3, I argue that the challenge should be answered by appealing to the essence of *to be necessary*. In §4, I will clarify that my suggestion is legitimate, especially that it is not viciously circular. My main thesis of this paper is merely conditional: if one is committed to essentialism, one must also hold that *to be necessary* has essentialist nature.

1 Introduction

By “necessity” I mean *metaphysical* necessity if not otherwise stated. It is further supposed that all logically necessary or conceptually necessary facts are also metaphysically necessary. Similarly, it may be helpful to note that if there is a variety of grounding, by “grounding” I mean *metaphysical* grounding. By “essence”, I mean *mediate and consequential* essence, following Fine (1995a). Suppose we have a clear idea of a strict and direct essence, which is immediate and constitutive. Then, mediate essence of F includes both the nature of itself and the nature of things that are involved in the nature of F ; consequential essence of F includes what logically follows from the nature of F .

Also crucial is the distinction between objectual essence and generic essence. On the one side there is objectual essence, the essence of a particular object, and the other side the essence of *to be* such and such, that is, generic essence. Fine (2015) argued the essence of *to be* F is different from the essence of the concept F . Suppose the objectual essence of the concept *concept* includes the fact that *concept* is a concept. Then it is a consequential essence of it that $\exists x(x \text{ is a concept})$. However, it seems quite implausible to think that it is by nature of what it is to be a concept that there is something satisfies it. That is, while $\exists x(x \text{ is a concept})$ is plausibly essential to the concept *concept*, it would not be essential of *to be a concept*. This, we may say, illustrates the difference between the objectual essence of concept and generic essence of *to be a concept*. It would be clear that the essence of objects or things would be the objectual essence of that object.

I take the primary subjects of being essential, as well of being necessary to be facts (which not necessarily actually obtain), and take grounding to be the higher-order relation between facts. Yet it will not be very difficult for whom favor propositions, or first-order relational view of grounding translate the words used in this paper to their favored account. Sometimes I will use “an essence of F ” to denote a fact essential to F , whereas “nature” is used strictly for the non-further-entity sense of essence (Lowe 2008).

I do *not* use two terms “be essential of/to” and “obtain in virtue of the nature of”(equivalent with “obtain essentially of”) interchangeably, since I operate with non-factive notion of essence, where facts essential to whatever may not obtain. If a fact obtains essentially of something, it is essential to the thing in question; but not vice versa. Following Fine (2015), by $p \leftarrow_{x_1, \dots, x_n} F(x_1, \dots, x_n)$ and $q \leftarrow r$ I mean p is a generic (mediate and consequential) essence of F and q is a generic essence of

r respectively. Objectual essence will be notated by using a *rigid predicate* $\lambda x.(x = a \vee x = b \vee \dots)$, where $a, b, \dots$ are names for object. To say p is an objectual essence of a I write “ $p \leftarrow_x (\lambda y.y = a)x$ ”. For the notion of obtaining essentially of something, I borrow from the traditional notation (Fine, 1995a) and write “ $E_F p$ ” for p obtains in virtue of the nature of F , where “ F ” is possibly 0-placed. By convention, I sometimes write “ $E_a p$ ” as an abbreviation of “ $E_{\lambda x.x=a} p$ ”.

By “essence fact” I mean the fact that whatever fact is essential of whatever. Furthermore, I assume that if whatever is essential of F , such an essence fact obtains essentially of F . That is, if $p \leftarrow_{x_1, \dots, x_n} F(x_1, \dots, x_n)$, then $E_F(p \leftarrow_{x_1, \dots, x_n} F(x_1, \dots, x_n))$. By “realized essence fact” I mean the fact that whatever fact obtains essentially of whatever, i.e., the fact that $E_F p$ for some F . I also assume that a realized essence fact obtains essentially: $E_F p \supset E_F E_F p$ where “ $\supset$ ” is used for material conditional to prevent ambiguity

By “essentialism”, I denote the thesis that for any necessary fact p , $\Box p$ is fully grounded by the fact that p obtains essentially of whatever, i.e., $E_F p$ for some F . This can be regarded as a proposed answer of accounting the ultimate metaphysical source of necessity (Hale, 2002). The source of necessity is supposed to provide the metaphysical explanation of why a necessary fact is necessary. It does not seem likely that it must always also explain why the fact obtains. Sometimes that p is necessary may be partially explained by that p obtains. It is customary to explain the necessity of the identity between Hesperus and Phosphorus by the actual identity of them, and the necessity of identity¹. In this case, the particular source for the necessity of p would include p , which would be inadequate to explain p itself. However, this cannot be the *ultimate* source for necessity, for in explaining the necessity of the fact that Hesperus is Phosphorus we appealed to the necessity of identity. We might further ask why, then, every identity between an object and itself is necessary; and if it is again explained via appealing to the necessity of another fact q , we might ask why q is necessary, and so on.

The chain of explanation might continue infinitely, or end at the point where we can give no further explanation for the necessity in question, i.e., where the necessity is primitive. Or, the chain will end if one can give the explanation for the necessity in question with non-modal source. The final non-modal destination would be, then, the *ultimate* source of the particular necessity which was initially at stake.

¹Kripke (1980). Although it might be disputable whether this is a *metaphysical* explanation, at least presumably there is no quick argument against it.

Essentialism as I understand it, then, consists of two theses: that the ultimate source of *all* necessities are realized essence facts, and the source of necessity is the ground of the necessity. The first thesis is particularly strong. Some philosophers favor a weaker view: that some necessities have their ultimate sources on essence facts, but some other necessities do not². In this paper, I focus on the more ambitious essentialism which states all necessities are grounded in some realized essence facts.

The grounding thesis is explicitly endorsed by some philosophers³. Correia (2024) suggests that the grounding thesis is not forced upon theorists who wish to seek the ultimate source of necessity in essence. He suggests that instead of grounding, one might hold that necessity is reduced, or defined in terms of essence (as in Fine (1994) and Fine (1995a)), or there is a generic identity between essence facts and necessities (as in Correia and Skiles (2021)). I admit that these are real possibilities, and by “essentialism” I *exclude* accounts that do not involve the grounding thesis⁴. But it should be noted that there indeed are cases where grounding thesis is implicit. For example, Kit Fine, when he defines *to be necessary* in terms of *to be essential of all objects*, he would have real constitutive definition in mind, which gives both grounding and essence claim (Fine, 2015).

Finally, some comments should be given for the restriction on *objectual* essence which can be found occasionally. This qualification is explicit in some writings, for example Fine (1994): “The metaphysically necessary truths can then be identified with the propositions which are true in virtue of the nature of all *objects* whatever” (p.9, italics added); Hale (2013): “Metaphysical necessities might plausibly be identified as those which hold in virtue of the nature of *things* in general.” (p.150, italics added). The general picture then would be that for any necessary fact p , we can find a rigid predicate ρ such that $E_\rho p$ which grounds $\Box p$.⁵

In paradigms where essence of to be whatever is treated as a constitutive necessary condition (Fine,

²For example Hale (2013, Ch.6) holds that the necessity of objectual essence facts are primitive, and Godman, Mallozzi, and Papineau (2020), Correia and Skiles (2021) doubt that the necessity of logical truths have essentialist source.

³For example Lowe (2012), Hale (2013, Ch.6), and Wallner and Vaidya (2020)

⁴For instance the account which holds that necessary facts and essential facts are extensionally equivalent, yet necessity is ungrounded (Wilsch, 2017). Similarly, Correia and Skiles (2021) hold that *to be necessary* is generically identical with *to be a logical consequence of a true generalized identity* (or its extensional correlate), thus the relation between them is symmetric, while grounding is thought to be asymmetric.

⁵Strictly speaking, in the logic proposed by Fine (1995a), for any predicate F not necessarily rigid, given $E_F p$ we can find a rigid predicate ρ such that $E_\rho p$. This is due to the axiom (II) (v): $F \subseteq G \supset (E_F A \supset E_G A)$. Intuitively, it says that if the intension of F is a subset of the intension of G , any essence of F is also an essence of G . Thus, given the rigid predicate of being one of all objects $\vee$, if $E_F p$, we could get $E_{\vee} p$. In this case, it seems to say for any necessary fact p there is an essence fact such that $E_F p$ is at least extensionally equivalent to the original thesis proposed by Fine. However, the same axiom yields the metaphysically undesirable result that for any two intensionally equivalent predicate F and G $E_F p \Leftrightarrow E_G p$, regardless of their hyperintensional difference. Thus, I do not endorse the logic of Fine (1995a) throughout the discussion.

2015), or as something generically identical with it (Correia & Skiles, 2017), essences are straightforwardly not always necessary. For example, let “ sg ” stand for the fact that *snow is green*. Then it is extremely plausible that $sg \leftarrow sg$; yet certainly it is not necessary that snow is green. After all, it does not even actually obtain. This might seem to provide a motivation for the restriction on objectual essence, since we would not like to grant $E_{sg}sg$ and thus license $\Box sg$. However, $sg \leftarrow sg$ simply does not grant $E_{sg}sg$ in our sense. As I mentioned earlier, the *essence fact* about F – not the essence of F – obtains essentially of F . Thus, $sg \leftarrow sg$ licenses only $E_{sg}(sg \leftarrow sg)$; and as Fine (2015, p.299) suggested, $p \leftarrow q$ yields $q \supset p$, thus it would mediate and consequentially yield $E_{sg}(sg \supset sg)$ ⁶. This is acceptable, since we would have no problem accepting $\Box(sg \supset sg)$: necessarily, snow is green if snow is green. I would like to take one step further and protest that $E_{sg}(sg \leftarrow sg)$ is *grounded* on $sg \leftarrow sg$, yet this is optional and perhaps contentious.

Thus, I see no particular reason for essentialists to think only objectual realized essence facts can ground necessity. It is even unclear how would an essentialist account for the necessity such as *all bachelors are unmarried* without appealing to the generic essence of *to be a bachelor*. By contrast, let “ U ” be the predicate “*to be unmarried*” and “ B ” “*to be a bachelor*”, with generic essence we can have $Ux \leftarrow_x Bx$, which yields $E_B \forall x (Bx \supset Ux)$, which could ground $\Box \forall x (Bx \supset Ux)$, i.e., it is necessary that all bachelors are unmarried. With such consideration, I will discard the restriction on *objectual* essence throughout the discussion. Still, it is worth noting that all objectual essence *regarding itself* obtain essentially of the object in question, in the mediate and consequential sense. For example, suppose $Fx \leftarrow_x (\lambda y.y = a)x$. This yields $E_{\lambda y.y=a} \forall x ((\lambda y.y = a)x \supset Fx)$, and by β -reduction $E_{\lambda y.y=a} \forall x (x = a \supset Fx)$, and by the rule of identity $E_{\lambda y.y=a} Fa$.

2 The Challenge of Gap

Obviously, all essentialist accounts should not take *to obtain essentially* to be understood in terms of *to be necessary*. For example, suppose for Fa to obtain essentially of a just is for a to necessarily have F . Then to say the necessity of Fa is explained by Fa obtaining essentially just amounts to say Fa is necessary because a necessary has F . In this case essentialism results in blatant circularity.

⁶In general, $p \leftarrow_{x_1, \dots, x_n} F(x_1, \dots, x_n)$ would yield $\forall x_1 \dots \forall x_n (F(x_1, \dots, x_n) \supset p)$.

On the other hand, it is the thesis of essentialism that for any F and p , $E_F p$ grounds $\Box p$. But then one might ask if essence should not be understood in terms of modality, i.e., has a non-modal nature, how can one ensure such a close relationship between *to obtain essentially* and *to be necessary*. The challenge is often articulated in terms of extensional relation between them. The grounding thesis implies what Mackie (2020) calls “Necessity Principle (NP)”:

$$(NP) \quad \forall p(E_F p \supset \Box p)^7$$

and it is challenged that if essence is non-modal, essentialists cannot even account for why (NP) holds. Of course, essentialists may contend that (NP) obviously holds. However, the problem I want to discuss here is not epistemic but metaphysical⁸. Even if essentialism is correct, (NP) would not be a fundamental fact of the reality⁹. There seem at least two reasons for essentialists to believe the non-fundamentality of (NP). First, since (NP) involves universal quantification, it is plausibly grounded on each instances of p satisfying $E_F p \supset \Box p$, which is plausibly further grounded on either $\neg E_F p$ or $\Box p$. Second, essentialism implies that *to be necessary* is grounded, and therefore nonfundamental; whereas if (NP) is fundamental, the constituent of (NP) including *to be necessary* (viz., ‘ $\Box$ ’) is fundamental, thus contradiction.

Thus, it would make sense to ask for the metaphysical explanation for (NP). But then it is unclear how non-modal conception of essence can give metaphysical explanation of why (NP) holds. (NP) would have a solid metaphysical basis if *to obtain essentially* is simply defined in terms of necessity. Yet as we saw before, this is not something essentialists can admit. The predicament can be illustrated as follows. To avoid circularity, essentialists should not take essence, or more specifically *to obtain essentially* to have modal nature. But to provide metaphysical basis for (NP), it seems that they should think there is a modal import in essence. But it is impossible to say essence is both non-modal and modal. As Mackie writes: how can essentialists “deliver a modal rabbit out of a non-modal hat”? (Mackie, 2020, p.252, also see Leech, 2018 and Romero, 2019)

Preexisting attempts by essentialists to provide metaphysical basis for (NP) faced strong objections. One common understanding of essence among essentialists is by *real definition*: roughly, it is the

⁷where “ F ” is any n -place predicate for any $n \in \mathbb{N}_0$

⁸Correia and Skiles (2021) gave an excellent argument in reply to the epistemic challenge, and it seems that the premise of the argument can be weakened so that any essentialists who hold essence is intimately related to identity (not necessarily defined in terms of identity) can utilize.

⁹I assume that only ungrounded facts can be fundamental, that is, the ‘minimalist’ worldview of Fine (2001).

thesis that the constitutive and immediate essence of certain thing or *to be such and such* is captured by their real definition¹⁰. Hence, if some x lack the essential property of an object α , it means that it fails to satisfy the definition of α , thus we can conclude that $x \neq \alpha$. Hale tried to argue that this conception of essence already grants (NP). Suppose $\Phi\alpha$ is an objectual essence of α . Now suppose it is possible that $\neg\Phi\alpha$. It implies that there is a β such that $\alpha = \beta \wedge \neg\Box\Phi\beta$. But β cannot be identical to α in the world such that $\neg\Phi\beta$. For, it does not satisfy the condition to be identical to α , which is Φ . Therefore by the necessity of identity $\neg\alpha = \beta$, contradiction. Since contradictions are impossible, we can conclude that it is not possible that $\neg\Phi\alpha$. In other words, if $E_{\lambda x.x=\alpha}\Phi\alpha$, then $\Box\Phi\alpha$ (Hale 2013, p.133).

But it has been pointed out that definitional conception of essence cannot guarantee (NP) at least in the way Hale proposed (Romero 2019, pp.125-7; Leech 2021, pp.893-5). For the proposed argument to succeed, it must be assumed that Φ is *necessarily* the definition of α . That is, either $\Box E_{\alpha}\Phi\alpha$ or $\Box\Phi\alpha$ is required. But what is assumed is only that Φ is *actually* the condition of the identity. And to hold that the assumption $E_{\alpha}\Phi\alpha$ implies $\Box\Phi\alpha$ simply begs the question.¹¹

Alternatively, Correia and Skiles (2017) suggested that essence should be understood in terms of generalized identity. Let “ $\phi \equiv \psi$ ” mean that ϕ is generically identical to ψ . Then q is a full essence of p iff $p \equiv q$, and r is a partial essence of p iff there is a q such that $p \equiv (r \wedge q)$ (pp.649-50). And to understand essence in terms of generalized identity may seem to offer better prospects for advocating essentialism¹², since “[a]s with objectual identity, generalized identity is tightly linked to metaphysical necessity” (Correia and Skiles 2017, p.656). Here $p \equiv p$ is a theorem¹³ and therefore in any normal modal logic it follows that $\Box(p \equiv p)$. And given $p \equiv (q \wedge r)$, which means that q is a partial essence of p , by substitutability we get $\Box(p \equiv (q \wedge r))$. That is, assuming q is a partial essence of p , it follows that q is a partial essence of p *necessarily*.

This actually proves something stronger than (NP): namely, $E_F p \supset \Box E_F p$. However, since anything obtains essentially of whatever would obtain *simpliciter*, this consequence implies (NP). The

¹⁰See, for example Fine (1994), Lowe (2012), or Hale (2013)

¹¹Of course, Hale himself would wish to reply that the necessity of $E_{\alpha}\Phi\alpha$ is not to be explained by further essence. Rather, it is a fundamental necessity that do not permit further explanation (Hale 2013, p.158). However, this move, whether it is convincing or not, is not permitted to essentialists in our sense.

¹²As far as I know, there are no actual essentialists who argue for essentialism in our narrow sense with such a conception of essence. The allegedly failed argument below is rather constructed by challengers of essentialism (Romero, 2019; Leech, 2021).

¹³For they take generalized identity to be reflexive (p.645)

problem here is that the proof presupposes $\Box(p \equiv p)$: that the generalized identities are necessary. Then what could be the source of the necessity of $(p \equiv p)$? It has been argued that the necessity of $\Box(p \equiv p)$ cannot be explained by the essence of *to be generically identical* (Leech 2021, pp.901-2; cf. Romero 2019, p.138). For example, suppose that the necessity is explained by that *for two things to be generically identical is partially to be in a reflexive relation*. Grant, but what explains the necessity of the partial generalized identity between generalized identity and reflexive relation? Here we cannot appeal to the necessity of generalized identity, for that exactly is the supposed explanandum. Therefore, to appeal to the essence of generalized identity does not solve the problem.

Leech seems to conclude that in explaining the necessity of the generalized identity “*one should not appeal to the essence of anything to account for this necessity*”. She further considers this constraint as “the crucial and general objection” on the necessity of facts obtaining essentially (p. 902).

I disagree. I think that Leech overlooks a remaining possibility: that all generalized identities are necessary because of the essence of *to be necessary*. Furthermore, in my view, this also applies directly to (NP). I agree that any non-modal specification of essence cannot be used to metaphysically guarantee (NP). However, every fact obtaining essentially is necessary, not because of the nature of what it is to obtain essentially but because of the nature of what it is to be necessary. In the next section, I first argue that essentialists must take (NP) obtain essentially of *to be necessary*, and show how adopting this thesis help essentialists respond to the challenge of gap.

3 (NP) as a Fact Obtaining Essentially of *to be Necessary*

Essentialists who take (NP) to hold would also wish to take (NP) to hold *necessarily*. There could be various motivations behind. If an essentialist takes side on *grounding necessitism*, viz., the thesis that the grounds necessitates the grounded, it would be a natural consequence that (NP) is necessary. Due to the popularity of necessitism, this might already provide good reason for essentialists in general to adopt the necessity of (NP)¹⁴. But even if one favors contingentism, viz., the thesis that the grounds do not necessitate the grounded, may still take (NP) to hold necessarily in order to avoid conflict with plausible metaphysical assumptions. To reason in terms of modal logic without giving metaphysical

¹⁴Furthermore, necessitism is extremely natural for essentialists who adopt a ‘unified foundation’ of essence and ground.

import, if (NP) is not necessary (in the actual world @) there is a possible world accessible from @ where $\exists p(E_F p \wedge \neg \Box p)$. And under essentialism the witness of p either does not obtain essentially in @ or it does and therefore $\Box p$ holds in @. In the latter case, we straightforwardly get a counterexample for S4 principle: $\Box p \supset \Box \Box p$, which is commonly thought to hold for metaphysical modality. In the former case, since essentialists also hold that $E_F p \supset \Box E_F p$ ¹⁵ it follows that all realized essence facts obtaining in @ obtain in all possible worlds; but in some non-actual possible worlds there obtain additional realized essence facts. But there do not seem to be a good reason to think that the actual world has the minimal realized essence facts among all possible worlds, if there can be a discrepancy of realized essence facts between worlds at all.

Thus, essentialists would admit that (NP) is necessary. And given essentialism, to say that (NP) is necessary amounts to say that (NP) obtains essentially of whatever. But of what would it obtain essentially, if it does at all?

It might be more promising to search for the subject which (NP) obtains essentially of by inspecting (NP) itself. Notice the formalization of (NP) is articulated by a universal quantification over facts, an operator E which reads “to obtain essentially of”, a material conditional, and a necessity operator $\Box$ which reads “to be necessary”. But it would be obvious that (NP) would have nothing to do with the nature of the material conditional, so we are left with three candidates.¹⁶ It might seem equally farfetched to think that (NP) obtains essentially of the logical object which is associated with the universal quantifier over facts. It might sound better to say (NP) obtains essentially of *all facts*. The general idea might be that, for each individual fact p it obtains essentially of p that $E_F p \supset \Box p$; and the totality of facts, if somehow given that the totality of facts contains *all facts*, it would obtain essentially of the totality of facts that for every fact, if it obtains essentially then it is necessary, i.e., (NP).

But could (NP) obtain essentially of yet something else? Well, trivially, if (NP) obtains essentially of one of three candidates mentioned in the previous paragraph, it will also obtain essentially of anything which their essence involves it. But let us exclude such cases into our consideration due to their metaphysical insignificance. Still, it seems one possible line of thought is that,

¹⁵Immediate from $E_F p \supset E_F E_F p$ and (NP).

¹⁶One might insist that I should also consider a polymorphic existential quantifier over n -place predicate for $n \in \mathbb{N}_0$. But it is contentious whether such a quantifier is desirable for the object-language; and even if it is desirable, I do not think it makes sense to think (NP) obtains essentially of whatever that object-language term is associated with. At least I fail to make sense of such an idea.

although (NP) is not fundamental itself, it might obtain essentially of fundamental metaphysical laws, or something similar.

In sum, we are left with four suggestions: either (NP) obtains essentially of *to obtain essentially*, of all facts, or of some fundamental metaphysical laws, or obtains essentially of *to be necessary*. I argue in detail that anything except the final option fails.

3.1 Taking (NP) to obtain essentially of *to obtain essentially*

It is worth noting that it is already suggested by Wilsch (2017) and Wallner and Vaidya (2020) that (NP) or something stronger than that obtains essentially of *to obtain essentially*, although the former would not count as essentialism in our sense. Wilsch directly takes (NP) to be the essence of essence, whereas Wallner and Vaidya takes both $E_F p \supset \Box E_F p$ and $E_F p$ can explain $\Box p$ essential to the operator E . Since $E_F p \supset \Box E_F p$ entails (NP), it would be fair to say both takes (NP) as a mediate and consequential essence - which I simply say ‘essence’ - of *to obtain essentially*.¹⁷

In response, Romero (2023) argued that this results in the claim that $p \wedge q$ grounds p which infringes the irreflexivity¹⁸. His argument is convincing if one adopts the conception of essence in terms of generalized identity. Suppose that $E_F p \equiv_p (\Box p \wedge q)$: to obtain essentially just is to be necessary and q . This would yield that $E_E \forall p (E_F p \supset (\Box p \wedge q))$ and consequentially $E_E \forall p (E_F p \supset \Box p)$ if we follow Correia and Skiles (2017). Now, since two generically identical statements would be substitutable even in hyperintensional contexts (Dorr, 2016; also see Rayo, 2013), if p grounds q , r such that $p \equiv r$ would also ground q . A brief example: suppose x ’s not being the subject of the marriage loan is grounded by x ’s being a bachelor. Since to be a bachelor just is to be an unmarried adult male, x ’s not being the subject of the marriage loan would also be grounded by x ’s being an unmarried adult male. Similarly, given $E_F(p) \equiv_p \Box p \wedge q$, the essentialist thesis that $E_F(p)$ grounds $\Box p$ is equivalent

¹⁷Wallner and Vaidya explicitly endorses that “modality is grounded in definitional essences.” (Wallner and Vaidya 2020, p. 431). They also make it explicit that they intend to follow Hale in giving non-transmissive explanation for the source of necessity in terms of ground. However, in case of Wilsch, he explicitly states himself as a modal primitivist, not an essentialist. Although he takes essence to give the “source” of metaphysical necessity, here “[t]he sources of necessity are the modal joints of nature.” (Wilsch 2017, p.429), which determine the extension of various kinds of necessity and account for the naturalness of them. Essence, then, explains why metaphysical necessity is a significant kind of primitive necessity, why it “carves the nature at its joints”. This is apparently a different enterprise from the essentialism being discussed in this paper.

¹⁸To be precise, he appeals to the irreflexivity of *metaphysical explanation* instead of grounding, since Wilsch is not committed to the thesis that essence grounds necessity. I focus on grounding to keep the discussion compact.

to the claim that $\Box p \wedge q$ grounds $\Box p$. And that apparently does not seem satisfactory, therefore the supposition is inconsistent with essentialism.

However, it is not obvious whether this argument works if one do *not* endorse Correia and Skiles' view on essence. The argument crucially relies on the premise that if q is the full essence of p , " q " must be able to substitute " p " in grounding context. This is not intuitively obvious for two reasons. First, it is not very clear there is a metaphysically significant notion associated with "full essence". Second, even we grant the notion of full essence, one can still contend that the full essence of p , say q , is intimately related to, but yet not identical to p . Then we would at least require a substantial theory of essence and ground to contend that if p grounds r , q also grounds r .

But there is a general objection one can raise for essentialist who holds (NP) obtains essentially of *to obtain essentially*. Let us first introduce the notion of essential dependence: x essentially depends on y , in Fine's term: "if y is a constituent of a proposition that [obtains] in virtue of the identity of x " (Fine 1995b, p.275; the term "essential dependence" follows Correia (2008)). Therefore from $E_E \forall p (E_F p \rightarrow \Box p)$, it follows that *to obtain essentially* depends on the constituent $\Box$, that is, *to be necessary*. However, since essentialists also hold that $E_F p$ generically grounds $\Box p$, they also hold that *to obtain essentially* does not depend on *to be necessary*, contradiction. Of course essential dependency is not antisymmetric. However, what essentialists need is that *to be C* is strictly prior¹⁹, since standard understandings of ground (cf. Schaffer 2009, Fine 2012) take the ground strictly prior, or more fundamental to the grounded. Thus, essentialism and the thesis that (NP) obtains essentially of *to obtain essentially* are incompatible.

However, the appeal to essential dependence may seem somewhat awkward, for I am operating with the consequential sense of essence, whereas the notion of essential dependence is usually formulated in terms of constitutive essence for good reasons. For example, assume the standard definition of biconditional ' $\Leftrightarrow$ ' can be also read as a generic grounding claim: $((p \supset q) \wedge (q \supset p)) \rightarrow (p \Leftrightarrow q)$. The obvious implication would be that the higher-order relation $\supset$ is strictly prior to the higher-order relation $\Leftrightarrow$. Yet since $p \Leftrightarrow p$ is a logical truth, it would be a consequential essence of anything, including $\supset$. And if consequential essence also yields essential dependence, it turns out that $\supset$ depends

¹⁹Following Fine (1995b), I use the term "dependency" for the fundamentality relation that does *not* imply antisymmetry (p.282-3). If X depends on Y , one only can infer that X is *not prior* than Y , for the possibility of Y being dependent on X is still open. The term "prior to" is reserved for the antisymmetric fundamentality relation.

on $\Leftrightarrow$, contradiction.²⁰

Now, if (NP) is proposed as obtaining constitutively essential of *to obtain essentially*, the aforementioned worry does not occur - and my argument still works fairly well. And even in some case where it is constitutive it may not affect the validity of the argument. For example, essentialists in fact must accept not only the conditional but also the biconditional $\forall p(E_F p \Leftrightarrow \Box p)$. Then given the motivation, one might want to say it is in fact the biconditional which obtains essentially of *to obtain essentially*, and (NP) is merely consequential. I find this proposal reasonable. Nevertheless, my argument *still* works fairly well, for “ $\Box$ ” occurs in the biconditional as well.

So we must examine *how* (NP) would obtain consequentially essentially. The following, although not mutually exclusive, exhausts the logical possibility: (1) (NP) follows from logic alone; (2) the set of facts obtain essentially of *to obtain essentially* is inconsistent; (3) $\forall p \neg E_F(p)$ follows from what obtains constitutively essentially; (4) $\forall p \Box p$ follows from what obtains constitutively essentially; (5) constitutive essence includes $\Box$; (6) constitutive essence includes $\Diamond$.²¹

(1) – (4) are untenable, whereas (5) and (6) shows that *to obtain essentially* still depends on *to be necessary* or something similar. (1). It would be clear that (NP) does not follow from some customary logics - classical first-order logic, normal modal logic, and so on. If (NP) is a logical fact in that sense, the challenge of gap would not arise at all, for we do not generally ask for the metaphysical explanation for a logical truth even the explanation can be given. Of course, we might construct a logic where (NP) follows logically. But then we will be confronted with the problem of philosophy of logic that why should we adopt such a logic, and with high chance the philosophical motivation would be essentialism itself. Then the opponent can raise the challenge of gap again, which will bring us back here once again. Therefore, one cannot appeal to (1) to escape the priority problem.

(2). Since what obtains in virtue of the nature of what would obtain *simpliciter*, taking (2) means taking the world to be inconsistent. I do not think this is a price many philosophers want to pay, and those who willing to pay the price would typically refuse to say we can infer everything from incon-

²⁰Fine (1995b) tries to solve this problem by introducing the additional condition: “ x depends upon y just in case y cannot be generalized out of the consequentialist essence of x ” (p.278), where y can be ‘generalized out’ of the essence of something if all occurrence of y in the essence can be substituted by any object and still belongs to the essence. This perhaps can solve the problem regarding the dependence between *objects*. Nevertheless, it does not seem to work regarding dependence outside first-order objects. What I proposed seems to be a counterexample: $\Leftrightarrow$ in the consequential essence of $\supset$ cannot be generalized out, for substituting into some two-place relation over propositions or facts would not result in logical truths. Also see Vogt (in press)

²¹I have classical S5 logic in mind; but it is also exhaustive even if one endorses S4 or intuitionist logic.

sistent premises. (3). Together with essentialism this means that there are no sources of necessity, and therefore no necessary facts. It is intuitively absurd, and I do not think the task of making sense of the proposal and refuting it would be adequate for the purpose of the paper. At least dialectically, I expect one who enters the debate of the source of necessity would not hold there are no necessary facts. The same consideration applies to (4), which means that all facts are necessary.

(5). It would be obvious that if the constitutive essence of *to obtain essentially* includes $\Box$ ²² it follows that *to obtain essentially* indeed depends upon *to be necessary*. And as I argued, this undermines the initial essentialist account. To show (6) still results in undesirable consequence is not straightforward. One easy way out, is of course to take $\Diamond$ depend on $\Box$, so that you cannot get (6) without (5). But perhaps one would prefer to take $\Box p$ to be defined as $\neg\Diamond\neg p$, or two operators to be on a par in regard of fundamentality. Throughout the paper, I implicitly assumed that $\Box$ is the fundamental modal operator, and essentialism is an account on the source of *necessity*, not other kinds of metaphysical modality. However, this assumption was made only because it facilitates the discussion. It is certainly possible for the essentialist account to be formulated in terms of possibility: the source of *possibility* of any possible fact q is explained by $\neg E_F(\neg q)$. If $\Diamond$ is more fundamental than $\Box$, essentialism probably should undergo such a modification. But once the modification is made, (6) undermines the account, for it implies that *to obtain essentially* is dependent upon *to be possible*.

Hence, taking (NP) obtain essentially of *to obtain essentially* would make it essentially dependent on *to be necessary*. Still it might be argued that this does not preclude *to obtain essentially* grounding *to be necessary*. The preclusion is by the following reasoning. (i) If *to be F* essentially depends on *to be G*, *to be F* is not prior to *to be G*; (ii) If *to be F* grounds *to be G*, *to be F* is prior to *to be G*. Conclusion: if *to be F* essentially depends on *to be G*, *to be F* cannot ground *to be G*. The criticism is that for the inference to be valid, “prior to” in (i) and (ii) should express the same metaphysical relation, which does not seem very likely (Vogt, 2026).

But to draw the conclusion one need not hold that (i) and (ii) operates with the identical notion of priority. To show that the priority cannot head towards opposite directions would suffice. This would be done, for example, if we show that both grounding and essence gives constitutive condition, where presumably constitution cannot be circular (Fine, 2015). Also, one can argue that the real definition - which is sometimes taken to be the constitutive and immediate essence - of whatever can

²²– which is the necessary condition of what obtain essentially of *to obtain essentially* includes $\Box$.

be formulated in terms of grounding (Rosen 2015, pp.198-9²³). I am inclined to think the account of Fine (2015) is largely correct, but to argue for the macroscopic metametaphysical theory would certainly exceed the scope of the paper. Still, some direct arguments for the opposite-heading of essence and ground should be dealt.

Vogt (2026) suggests that, assuming Aristotelian account of property, there can be cases where $E_a Pa$ but Pa grounds *property P exists* (p.519). Vogt seems to regard this being the case that a being both prior (by grounding) and not prior (by essential dependence) to P . However, it should be noticed that ' P ' is used once as a name for a first-order object, once as a predicate. Thus, what the grounding statement and the essence statement jointly imply is actually that a is not prior (by essential dependence) to *to be P*; where a is prior (by grounding) to the *property P*. This is clearly not a case of two priority relations heading the opposite direction.

Vogt (2026, in press) also cites various specific cases where one might contend that Fx grounds Gx but $E_F(Fx \supset Gx)$ or $E_{\lambda p, p=Fx}(Fx \supset Gx)$ (see also Correia (2018)). Of course, she is cautious enough not to declare any of the particular example to be definitive. However, it is suggested that the candidate examples, taken together, has enough force to make one to take such possibility seriously. I disagree, for it seems that, even though that the grounding claim and the essentiality of the conditional is correct, those putative examples are subject to a uniform treatment. That is: the conditional should obtain essentially of the grounded, not the ground. For example, Vogt holds in both paper that a *promises that p* grounds a *has the obligation to p*, whereas it seems that it obtains essentially of *to promise that* (one promises that $p \supset$ one has the obligation to p). One of course may deny that there is such an intimate connection between *to promise* and *to have obligation*. If there is, my treatment is to deny the essence claim, and alternatively say it obtains essentially of *to have obligation that* (one promises that $p \supset$ one has the obligation to p).²⁴²⁵

Therefore, insofar as essentialists tries to ground *to be necessary* - alternatively, *to be possible* - in

²³Notice that the final proposal in p.200 is strikingly similar with Fine (2015): a real definition giving a necessary and sufficient condition must involve both essence and ground.

²⁴This is exactly parallel to the goal of this section. Essentialist would agree that $E_F(p)$ grounds $\Box p$. But if one also have to think that the conditional $E_F(p) \rightarrow \Box p$ should be essential, then it should be essential of the grounded, that is, *to be necessary*, not of the ground, that is, *to obtain essentially*.

²⁵One might criticize such a treatment as *ad hoc*. It perhaps is in some sense, but notice this does not increase the theoretical complexity of priority, but reduces it. If the priority by grounding and priority by essential dependence was so fundamentally different so that it permits opposite-heading, it seems that there should be two fundamentally different account of what metaphysical priority is, perhaps even two different kinds of fundamental reality, which should be altogether compatible. That not being present, I would regard my move justified.

to obtain essentially, they should not think (NP) obtain by the nature of *to obtain essentially*.

3.2 Taking (NP) to obtain essentially of fundamental laws, or all facts

Extremely similar considerations would lead us to reject the suggestion that (NP) would obtain essentially of some fundamental laws. Recall that I argued that (NP) is not fundamental in §2. Then we can utilize again the notion of essential dependence, and argue that if (NP) obtain essentially of some fundamental laws, the laws in question depends on (NP), and therefore (NP) is prior than those laws or is as fundamental as them. But since nothing can be prior than the laws since they are assumed to be fundamental, (NP) would as fundamental as the fundamental laws, i.e., fundamental. This contradicts with the observation that (NP) would not be fundamental, especially if essentialism holds.

Again the argument in the previous paragraph is awkward since it operates with the consequential notion of essence. Still, the list (1)–(6) is still exhaustive even we substitute the subject of obtaining essentially of into the fundamental laws in question, and exactly same argument can apply. To repeat: let L be the (totality of) law which (NP) allegedly obtains essentially of. Given $E_L(\text{NP})$, either (1)' (NP) follows from logic alone, (2)' the set of facts obtain essentially of L is inconsistent; (3)' $\forall p \neg E_F(p)$ follows from what obtains constitutively essentially of L ; (4)' $\forall p \Box p$ follows from what obtains constitutively essentially of L ; (5)' constitutive essence includes $\Box$; (6)' constitutive essence includes $\Diamond$. (1)' is obviously not the case. (2)' implies that the actual world is inconsistent; and if one can get along with this consequence, one will not think (2)' would license (NP) to obtain essentially of L automatically. (3)' says nothing is necessary given essentialism, and (4)' straightforwardly says everything is necessary. They are implausible *per se*, and moreover anyone enters the debate of the source of necessity would not wish to commit to any of both theses. Finally, (5)' and (6)' yields the incompatibility between essentialism (or its suitably modified version) and $E_L(\text{NP})$.

And it is again the consideration of dependency that undermines the idea that (NP) might obtain essentially of *all* facts. The idea suggests that for each fact p , $E_F p \supset \Box p$ obtains essentially of p . Now suppose p is a fundamental fact. By essential dependence, it follows that $\Box$ is fundamental; whereas by the grounding thesis of essentialism it follows that $\Box$ is not fundamental.

One might criticize that it is unfair to make the assumption that if (NP) obtains essentially of all facts, every particular facts should have necessity-related essence. The problem I raised will not

occur if, only some nonfundamental facts have necessity-related essence, and these limited cases are sufficient to grant (NP) to obtain essentially of the totality of facts. I find such scenarios implausible. In those scenarios, not only selected nonfundamental facts should guarantee, by their nature, that for numerous facts, especially fundamental ones, if they obtain essentially then they are necessary; it should also guarantee that those facts are *all the facts out there*. I simply fail to imagine any putative cases where the latter can be achieved.

It may be helpful to clear out some potential confusion that might motivate the criticism above. It is indeed possible that (NP) is *essential* to some nonfundamental facts. In fact, I hold that (NP) is essential to (NP) itself! But we cannot infer that (NP) *obtain essentially* of whatever from it.

3.3 Taking (NP) to obtain essentially of *to be necessary*

The only remaining possibility for non-transmissivists, then, is to say that the relevant instances (GNP) is an essence of *to be necessary*: $E_{\Box} \forall p (C(p) \rightarrow \Box p)$. By doing so one can insist that *to be C* suffices for the full source of necessity. For in this case nothing else than the nature of *to be necessary* is required to guarantee that (GNP) holds, and therefore one need not posit a further source of necessity that validates the leap from $C(p)$ to $\Box p$. It also does not make *to be C* not prior to *to be necessary* and thus making the grounding claim suspicious. By placing (GNP) into the essence of *to be necessary*, it follows that *to be necessary* is not prior to the constituent of its essence: *to be C*, which is in harmony with the grounding claim that $C(p)$ grounds $\Box p$.

Since this is the only option left for non-transmissivists, they should take the relevant instance of the essence claim into the account. For example, possible world realists must say it is somehow the nature of *to be necessary* which guarantees that whatever true in all possible worlds are necessary. Applied to essentialism, the claim would be that it is by the essence of *to be necessary* that whatever is objectually necessary is necessary. Below I will abbreviate this claim as (NPEN) and the generalized version (GNPEN). That is:

$$(NPEN) \quad E_{\Box} \forall p (\exists F E_F p \rightarrow \Box p)$$

$$(GNPEN) \quad E_{\Box} \forall p (C(p) \rightarrow \Box p)$$

where again F rigid.

But how is endorsing (NPEN) different from Fine's original account? The apparent feature is that I do not appeal to the notion of essence of *all objects*, and postpone the problem what exactly should be the domain of the logic of essence. But it also provides essentialists the material to respond the challenge of gap which lacked in the definition. Following Fine (2015), we can reformulate his definition as $E_{\forall}p \leftrightarrow_p \Box p$. This grants, in older notation, that $E_{\Box}\forall p(\Box p \rightarrow E_{\forall}p)$, and since $\forall$ is rigid we trivially get $E_{\Box}\forall p(\Box p \rightarrow \exists F E_F p)$, F rigid. So Fine does grant that the converse of (NP) is essential of *to be necessary*, but it does not by itself grant (NPEN). Thus, although Fine's definition grants that *to be objectually essential* grounds *to be necessary*, it does not answer the challenge of gap - why should *to be objectually essential* must be such that it grounds *to be necessary*? By endorsing (NPEN), we now have enough material to reply to the challenge. It is misleading to ask whether *to be objectually essential* has a suitable nature to be the source of necessary. Rather, we must say that the nature of *to be necessary* is intimately related to *to be objectually essential*, and that provides the metaphysical basis for (NP) to hold.

Again, all essentialists must accept (NPEN). Otherwise, either essence turns out not to provide a full explanation for necessity; or essence cannot ground necessity because it infringes priority or irreflexivity constraint of ground. One consequence is that any compelling argument for essentialism would also be an argument for (NPEN). For I do not aim in this paper to argue for the essentialist answer against other alternatives, I will not explore possible specific arguments for (NPEN).

What I *should* show in this paper, is how essentialists can answer the challenge of gap by utilizing (NPEN). To briefly reiterate, their main criticism was that once we stick to the notion of essence that is *not* defined in terms of necessity, there seems to be a gap between essence and necessity. My suggestion is, of course: the gap is to be bridged by the essence of *to be necessary*.

Let us return to Hale's argument for the necessity of definitional essence. I take Romero's criticism to be correct that we cannot infer $\Box E_{\alpha}\Phi\alpha$ from $E_{\alpha}\Phi\alpha$ without appealing to the essence of *to be necessary*. To say $E_F p$ where F rigid is certainly not to say $\Box E_F p$. The gap, however, can be filled by the nature of what it is to be necessary, and an independent argument for the necessity of objectual essentiality can be given. By (NPEN), *T-axiom*²⁶ and modus ponens we get $\forall F\forall p(E_F p \rightarrow \Box p)$ where F rigid. By universal instantiation we get $E_F E_F p \rightarrow \Box E_F p$, and together with the theorem $E_F p \rightarrow E_F E_F p$ where F rigid, we can cut out the middleman and infer $E_F p \rightarrow \Box E_F p$, where F rigid.

²⁶ $E_F p \rightarrow p$ (Fine 1995a, pp.247-8)

Even better, we can revise Hale's argument in which does not depend on $E_F p \rightarrow \Box E_F p$. Suppose again, that $E_\alpha \Phi \alpha$. Now, Romero and Leech ask: why β such that $\neg \Box \Phi \beta$ cannot be identical to α , if the initial presupposition was only that it is *actually* $E_\alpha \Phi \alpha$? Now we have a further reply for that. Suppose for contradiction that $\alpha = \beta$ and therefore by substitutability $\neg \Box \Phi \alpha$.

Now, from (NPEN) we can prove, utilizing the *T-axiom* and modus ponens that $\forall F \forall A (E_F A \rightarrow \Box A)$. By universal instantiation we get $E_\alpha \Phi \alpha \rightarrow \Box \Phi \alpha$; and by the assumption $E_\alpha \Phi \alpha$ and modus ponens we get $\Box \Phi \alpha$. Contradiction. Therefore, there can be no β such that $\alpha = \beta \wedge \neg \Box \Phi \beta$, which ensures that if Φ is actually an essential property of α , then it is a metaphysically necessary condition for something to be identical to α .

Notice that this is nevertheless not to ignore the essentialist intuition that it is in some sense "obvious" that objectual essences are necessary. I suggested that essentialists cannot take objectual essence to be *ab initio* necessary. However, it seems also true that there certainly is some kind of frustration when essentialists are confronted with questions like "why are facts of objectual essential necessary?" or "why an object necessarily has the property that defines what the object is?"

This is, I suggest, same kind of frustration when one is confronted with questions like "why is water H_2O ?" or "why does Socrates' singleton set have Socrates as member?". As Glazier (2017) argues, we can simply answer: "that is just what water is" or "that is just what the set is". Turn to the question "Why are objectual essences necessary?", I suggest one kind of answer essentialists can give is: "that is just what to be necessary is!". And to give such an answer is to accept (NPEN).

4 Circularity

I argued in the previous section that the only viable option for non-transmissivists is to take the relevant instances of (GNP) to be a partial essence of to be necessary. This was done by showing that any other alternative is incompatible with non-transmissivism. But to say that it is the only viable option for non-transmissivists is not to say that it is tenable. After all, non-transmissivism might also be incompatible with (GNPEN); and with the argument I presented it completes a trilemma which likely undermines any non-transmissivist theory.

It seems clear that (GNPEN) is not *logically* inconsistent with non-transmissivism. To say what it

is to be necessary is such and such is one thing, and to say the necessity is explained by such and such factor is another. Similar to essentialism, which is an instance of non-transmissivism.

But it could be the case that the thesis seriously undermines the theoretical motivation of non-transmissivism. The reason of rejecting (GNP) being essential of *to be C* was not logical inconsistency. The argument against it was that it gives a verdict on priority which is inconsistent with non-transmissivism.

As I noted earlier, (NPEN) would not face the same problem. But perhaps there are other problems. Romero (2019) states six constraints on the notion of explanation which the ‘explanation’ of modality by essence must satisfy:

Why The explanation must answer to a ‘why’ question.

Acyclicity The explanation should not be circular.

Fundamentality The explanans should be metaphysically more fundamental than the explanandum.

Ontic and Factive It must be the case that the truth of the explanans, not our concept or theory of it, explains the truth of the explanandum.

Structural The explanandum need not be an object or a property.

Objectivity The explanation should be objectively valid. (p.124)

It is already shown that (NPEN) does not make essentialism infringe **Fundamentality**. Essentialism with (NPEN) also easily satisfies **Why, Ontic and Factive**. It answers the question “*Why is p necessary?*”, by appealing to the *facts* about essence of things, not our concept of essence. Essentialists would certainly hold that if the essentialist account on necessity is valid, then it is **objectively** valid. Finally, **Structural** is not a negative constraint that one can infringe.

However, whether essentialism with (NPEN) is compatible with **Acyclicity** is perhaps not so obvious. Some might feel uncomfortable to the idea that the nature of *to be necessary* validates the explanation from essence to *necessity*. But if it is indeed circular, what exact part of the picture is viciously circular?

Notice that I do not try to explain the necessity of p by (NPEN). The explanatory work is done by the fact that p is objectually essential. However, it is even unclear why involving (NPEN) in particular explanations would be viciously circular. Suppose I am wrong, and every explanation on necessity must include (NPEN). p is necessary because p is essential to some objects *and every such truths are necessary in virtue of the nature of to be necessary*. What is wrong with this kind of explanation? My point can perhaps be made by a simple analogy. If someone asks me why 3 is a prime number, I can simply answer “because 3 is a natural number that is not a product of two natural numbers smaller than itself.” However, if someone thinks this is incomplete, then I can add: “... *and every such natural numbers are prime in virtue of the nature of to be prime*”. It would be obvious that neither explanation for the fact that 3 is prime is viciously circular.

Perhaps another kind of charge can be made by asking how can one insist that (NP) is necessary. It might seem plausible to think that (NP) is also necessarily true, or at least essentialists must so insist. And if (NP) is necessary, by essentialism there would be some *object* n such that $E_{\lambda x.x=n}(\text{NP})$, hence $E_{\lambda x.x=n}(\text{NP})$ grounds $\Box(\text{NP})$. Persuasively the object n would be something relevant to necessity, and a further posit that the nature of n is grounded by the nature of *to be necessary* does not seem farfetched. But then $\Box(\text{NP})$ would be grounded by (NPEN), that is, $E_{\Box}(\text{NP})$. And this would seem somewhat problematic.²⁷

Firstly, it needs to be clarified whether essentialists are really committed to the necessity of (NP). I have formulated the account in a way that avoids such a commitment. Essentialists must hold that every fact essential to some *objects* is necessary, whereas my proposal was that (NP) is the *generic* essence of *to be necessary*. Thus, (NPEN) does not force essentialists to admit that (NP) is necessarily true. S4 principle: $\Box p \rightarrow \Box \Box p$ can also be derived without assuming (NP) is necessary, as long as one is willing to accept that (NP) is *logically* true in the modal logic of essence.

Nevertheless, suppose we somehow grant that (NP) is necessarily true, and $\Box(\text{NP})$ is grounded by (NPEN). Still, I do not see anything problematic here. To use another analogy: suppose a quasi-God (*‘Q’* below) knows all and only essential facts, and its knowing an essential fact is metaphysically grounded by that fact being essential. And suppose it is essential of *to know* that if someone knows that p , it is the case that p , that is, $E_K \forall p (\exists s K(s, p) \rightarrow p)$. It would then follow that $E_K \forall p (K(Q, p) \rightarrow p)$.

²⁷I believe this is what Jonathan Hricko meant when he suggested to me that if *to be necessary* has an essence, then necessity cannot be reduced to what is true in virtue of all essences. If it should be understood literally, the obvious reply is that I do not reduce necessity to what is essential, but to what is *objectually* essential.

Now, by the assumption on Q it would be the case that $K(Q, \forall p(K(Q, p) \rightarrow p))$, and it is partially grounded by $E_K \forall p(K(Q, p) \rightarrow p)$. It seems clear there is nothing viciously circular here. But notice that the proposed situation of $E_{\Box}(\text{NP})$ grounding $\Box(\text{NP})$ is structurally similar to the case of the quasi-God: there being a Φ containing X such that $E_X \Phi$, where $E_X \Phi$ grounds $X(\Phi)$. This does open the possibility that in some cases, *to be essential of to be X* can be prior to *to be X*. But this is not an obvious problem, for still the grounding is irreflexive. And if, through metametaphysical investigation philosophers somehow found out that *to be essential of to be X* cannot be prior to *to be X*, essentialists can simply insist that (NP) is not necessary.

5 Conclusion

I argued that essentialists, facing the problem of gap, must respond by accepting (NPEN) - that (NP) is true in virtue of the nature of to be necessary. This follows from the more general character of non-transmissivism. Not only essentialist theories on the source of necessity, but all non-transmissivist theories face the problem of the gap. When one asserts that p is necessary because $C(p)$, one has to elucidate the notion of C without modal notions, and at the same time grant (GNP): $\forall p(C(p) \rightarrow \Box p)$. I argued in section 3 that all non-transmissivists should regard (GNP) to be a partial essence of to be necessary. And since all essentialists theories are non-transmissivist - this is argued in section 1 - essentialists are forced to accept (NPEN). I also tried to show that this account is not circular as it may first appear.

What I did *not* argue in this paper is that essentialism is superior to its competitors. My aim in this paper has been to show that the particular challenge to essentialists can be met coherently without dismissing the challenge. The aim of the paper is defensive: although what the challenge from gap points out is correct, it is not strong enough to defeat essentialism. It rather helps essentialists to clarify what comes together with essentialism. *If* essentialism is correct, then so is (NPEN). Any argument in favor of essentialism is also an argument for (NPEN). Furthermore, if one agrees that *essence precedes existence* (Lowe, 2008), it illuminates another argumentative strategy for essentialists. A certain kind of necessity, existent or non-existent, is indeed specified by (NPEN). If essentialists succeeds in arguing *that* necessity is the metaphysical necessity, this consequence would strongly favor essentialism. But of course, the contrapositive of the conclusion also holds. If (NPEN) is

false, so is essentialism. Any criticism against (NPEN) would thereby be a criticism for essentialism. However, fleshing out new arguments for essentialism, or dealing with those criticisms would exceed the scope of this paper.

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